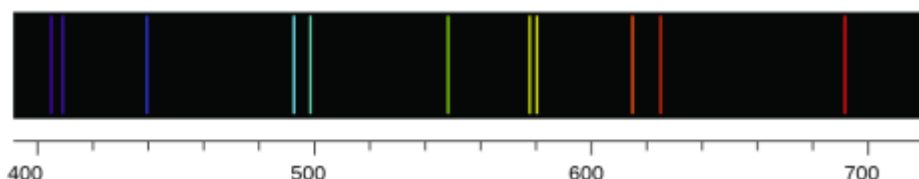


## Mercury spectrum- Determination of Wavelength and Photon energy

**Aim:-** To determine the wavelength and the corresponding photon energy of prominent lines of mercury spectrum

**Apparatus:** Spectrometer, grating, Mercury vapour lamp, reading lens and lamp.

**Principle:**



Diffraction grating a polished surface of glass having a large number of equidistant fine parallel grooves or slits (of the order of 600 lines/mm), used to diffract a composite light into its constituent colors.

Normal incidence means, the grating is mounted on the grating table, in such a way the incident ray falls on the grating at right angle.

Grating equation is  $d \sin \theta = n\lambda$ , where  $d$  = slit / groove spacing,  $n=1,2,3 \dots$

$$d = \frac{1}{N} = \text{Number of lines per unit length}$$

Wave length corresponding to a diffraction angle  $\theta$  is given by

$$\lambda = \frac{\sin \theta}{nN}$$

Where,

$N$  - number of lines per metre length of the given grating lines/ meter.

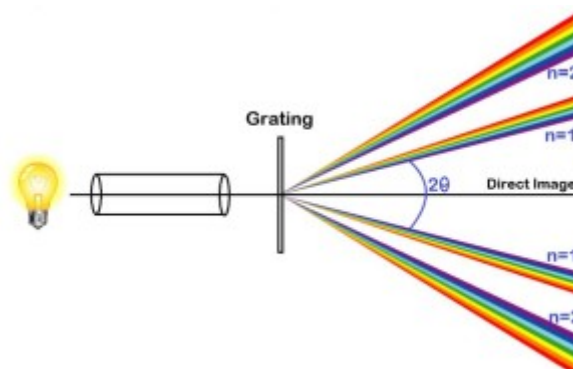
$n$  - order of the spectrum ( $n = 1$ , for the first order)

Photon energy associated with the wavelength  $\lambda$  is given by

$$E = hf = \frac{hc}{\lambda} = \frac{1240 \text{ eV} \cdot \text{nm}}{\lambda (\text{in nm})}$$

Thus energy of photon in eV is given by

$$E (\text{in eV}) = \frac{1240 \text{ eV} \cdot \text{nm}}{\lambda (\text{in nm})}$$



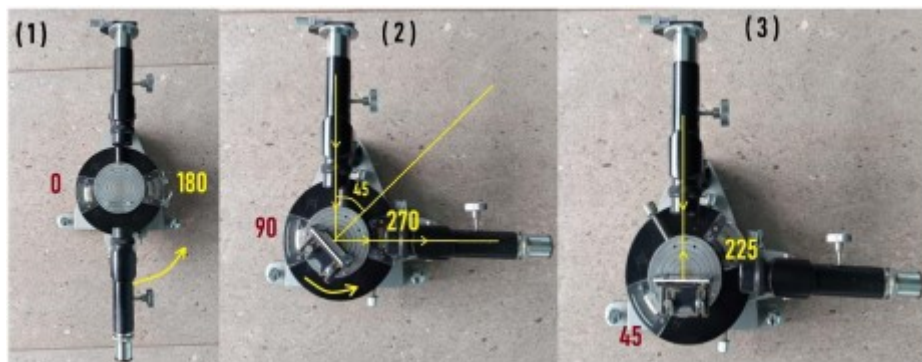
## Procedure:

### I- Initial Adjustments of the spectrometer

1. The telescope is focused towards a long distant object and the screw of the telescope is adjusted to get the clear image of the distant object. Now the telescope has been adjusted to receive the parallel rays.
2. The telescope is brought in line with the collimator and the well-defined image of the slit illuminated by a light source is obtained by adjusting the screw of the collimator. Make the collimator slit as thin as possible so that you see just a sharp thin line. The width of the slit image is made narrow.

### II. Setting the Grating for Normal Incidence

1. The telescope is brought along the direction of the collimator. The vertical crosswire to coincide with the image of the slit illuminated by the light source. Lock the telescope and then rotate the vernier disc until the reading on one of the scales become  $0^\circ$  (the reading of the other will be almost  $180^\circ$  )
2. Now the telescope is turned through  $90^\circ$  , then the grating table is rotated, such that the reflected image of the slit is made to coincide with the vertical cross wires of the telescope and lock the telescope and lock the prism table. At this position, vernier readings are  $90^\circ$  and  $270^\circ$  .
3. The Vernier disc is released and rotated by an angle of  $45^\circ$  in the appropriate direction such that the light coming out of the collimator will be incident normally on the grating. At this position vernier scale readings are  $45^\circ$  and  $225^\circ$  . Lock the prism table. Now the grating is fixed in the normal incidence position.



### III. To find the wavelengths of Hg Lines

1. The slit is illuminated by Mercury vapour lamp.
2. Release the telescope and rotate to the left side from the direct image position , to catch the first order spectrum .
3. The vertical cross-wire is made to coincide with various lines in the spectrum. Note the MSR and VSR readings of Vernier scales I and II, corresponding to each line.
4. Now the telescope is turned to the right side to catch the first order spectrum and the readings are taken as earlier
5. The difference between the left and right readings corresponding to a particular colour gives  $2\theta$  of that colour for the first order, where  $\theta$  is angle of diffraction.

**Observations:**

Number of lines per metre of the given grating  $N = 6,00,000$  lines/ meter.

Order of the spectrum,  $n = 1$ .

Least count of the spectrometer vernier

$$L \quad C = \frac{1}{n} \frac{MSD}{n} = \frac{0.5^0}{30} = \frac{30'}{30} = 1' \quad (1^0 = 60')$$

$$\text{Total Reading} = \text{MSR} + (\text{VSR} \times \text{LC})$$

| Line Colour | Vernier scale | Telescope position of the diffracted images |          |           | Difference $2\theta$ (2)-(1) | Mean $2\theta$ | Mean $\theta$ | $\lambda$ (nm)<br>$= \frac{\sin\theta}{nN}$ | $E$ (eV)<br>$= \frac{hc}{\lambda}$ |
|-------------|---------------|---------------------------------------------|----------|-----------|------------------------------|----------------|---------------|---------------------------------------------|------------------------------------|
|             |               |                                             | Left (1) | Right (2) |                              |                |               |                                             |                                    |
| Violet      | I             | MSR:<br>VSR:<br>TR:                         |          |           |                              |                |               |                                             |                                    |
|             | II            | MSR:<br>VSR:<br>TR:                         |          |           |                              |                |               |                                             |                                    |
| Blue        | I             | MSR:<br>VSR:<br>TR:                         |          |           |                              |                |               |                                             |                                    |
|             | II            | MSR:<br>VSR:<br>TR:                         |          |           |                              |                |               |                                             |                                    |
| Green       | I             | MSR:<br>VSR:<br>TR:                         |          |           |                              |                |               |                                             |                                    |
|             | II            | MSR:<br>VSR:<br>TR:                         |          |           |                              |                |               |                                             |                                    |
| Red         | I             | MSR:<br>VSR:<br>TR:                         |          |           |                              |                |               |                                             |                                    |
|             | II            | MSR:<br>VSR:<br>TR:                         |          |           |                              |                |               |                                             |                                    |

### Results:

The wavelength and the associated photon energy of mercury spectrum are determined and tabulated below:

[illegible]